

Red Hat Lightspeed 1-latest

Red Hat Lightspeed remediations guide

Fixing issues on Red Hat Enterprise Linux systems by using remediation plans

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Abstract

Create and execute remediation plans to remediate issues on any RHEL system registered with Red Hat Lightspeed.

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CHAPTER 1. REMEDIATIONS OVERVIEW

After identifying the highest remediation priorities in your Red Hat Enterprise Linux (RHEL) infrastructure, you can create and execute remediation plans to fix those issues.

1.1. ABOUT REMEDIATIONS

Remediations enables you to address **advisor recommendations**, **content advisories**, **vulnerability CVEs**, and **failed compliance rules found by Red Hat Lightspeed** on your connected RHEL systems.

You can remediate a single issue or a related group of issues by using a pathway in Red Hat Lightspeed. Pathways group multiple advisor recommendations under common actions for better efficiency.

For some issues, Red Hat Lightspeed provides several different remediation paths.

When you create a remediation plan, Red Hat Lightspeed generates an Ansible Playbook to implement the required remediation actions and apply any required patches on affected systems in your RHEL infrastructure.

Some issues require a manual fix and cannot be resolved by executing a remediation plan in Red Hat Lightspeed. To determine if you can remediate a problem in Red Hat Lightspeed, check the **Remediation type** value of the issue or recommendation.

Additional resources

- Remediating pathways

1.2. REMEDIATION TYPES

In Red Hat Lightspeed, an issue or recommendation for remediation can be either **Manual** or **Playbook**.

- **Manual:** Red Hat Lightspeed provides the manual remediation steps needed to fix or address all issues and recommendations, including whether the system requires a reboot for the remediation to take effect.
- **Playbook:** For many issues, Red Hat Lightspeed also provides a pre-built remediation playbook that automates the required resolution steps, which you can either:
- Run on your systems from within Red Hat Lightspeed
- Download and run externally in your Ansible Playbooks environment

1.3. RED HAT LIGHTSPEED REMEDIATIONS WORKFLOW

You can use the following outline of a remediations workflow to design how you create and execute a remediation plan.

- **Choose an issue or recommendation**
- Choose an issue or recommendation that Red Hat Lightspeed has detected on one or more of your RHEL systems.
- **Review the recommended resolution path**

- Determine which versions of RHEL are affected and whether or not a playbook is available. You can only create a remediation plan in Red Hat Lightspeed if a pre-built playbook exists.

- **Decide which RHEL systems to remediate**

- After you review the recommended resolution steps and determine if a playbook is available to remediate the issue, choose which systems to include in the plan.



IMPORTANT

To create a remediation plan for a group of systems, all systems in a group must be running the same RHEL major and minor versions. This ensures that the Red Hat Lightspeed-generated playbook resolution is compatible.

- **Create a remediation plan**

- The wizard in the Red Hat Lightspeed UI can help you create a remediation plan that you can access from the advisor, compliance, vulnerability, and patch service pages.
  To start the wizard and create a remediation plan, click **Plan remediation** after you select at least one system and one issue or recommendation for remediation. You can also create a remediation plan from the details page of a system, if Red Hat Lightspeed has detected issues that impact the system.

- **Decide how you want to execute your remediation plan**

- You can execute a remediation plan in Red Hat Lightspeed on directly connected Red Hat Enterprise Linux systems without additional subscriptions or tools. You can also download and run the associated playbook on your organization's Ansible Automation Platform (AAP) workflow.

1.4. PREREQUISITES FOR REMEDIATIONS

To create and execute remediation plans, you need a **subscription** and the appropriate **user access role**.

Subscription

Red Hat Lightspeed is included with every RHEL subscription. No additional subscriptions are required to use Red Hat Lightspeed remediation features.

User access role

By default, all Red Hat Lightspeed users automatically have access to read, create, and manage remediation plans.

- To remediate your Red Hat Enterprise Linux systems from Red Hat Lightspeed, you also need:
  - Access to Red Hat Lightspeed on the Red Hat Hybrid Cloud Console (Hybrid Cloud Console).
  - If you are using Red Hat Satellite, you need access to Satellite-managed systems on the console or in the Satellite application UI.
  - The required Hybrid Cloud Console User Access roles for managing and executing remediation plans.



IMPORTANT

All Red Hat Lightspeed users automatically have access to read, create, and manage remediation plans, but you need the **Remediations administrator** predefined User Access role to execute a remediation plan in Red Hat Lightspeed. Your **Organization Administrator** can grant User Access roles in Identity & Access Management settings on the Hybrid Cloud Console.

Remote host connectivity

To execute remediations, you must set up and enable the remote host configuration (rhc) within Red Hat Lightspeed. You also need to permit Red Hat Lightspeed users to execute remediation playbooks on rhc-connected systems, which can be done by enabling the Remote Host Configuration Manager (rhc) setting in Red Hat Lightspeed, provided you have the required administrative permissions.

Additional resources

- Remote Host Configuration Manager (rhc)

1.5. GET STARTED WITH REMEDIATIONS BY USING AN INTERACTIVE QUICK START

To help you get started with remediations, an interactive quick start is available in Red Hat Lightspeed on the Hybrid Cloud Console.

The Creating and executing remediation plans quick start guides you through the process in under 10 minutes and provides links to additional resources.

Prerequisites

- You must have a Red Hat Hybrid Cloud Console account and be subscribed to the Red Hat Lightspeed services.

Procedure

1. Log on to Red Hat Hybrid Cloud Console and then navigate to the Red Hat Lightspeed services.
2. To access the quick start, use one of the following steps:
   - **From within the Red Hat Lightspeed application:** Go to **Automation Toolkit > Remediation Plans**, and then click **Launch Quick Start** in the upper right corner.
   - **From the Quick starts library:** Go to RHEL Learning Resources.

1.6. MANAGE USER PERMISSIONS FOR RED HAT LIGHTSPEED SERVICES

Manage user permissions to control access to Red Hat Lightspeed applications. Use the User Access feature to apply role-based access control (RBAC). Red Hat provides predefined groups and roles to help Organization Administrators manage user permissions.

1.6.1. User Access overview

You can simplify access management in the Red Hat Hybrid Cloud Console by using role-based access control (RBAC). The User Access feature assigns specific permissions to roles instead of individual users, which are then inherited when roles are assigned to user groups.

You can also create custom groups and roles to provide more fine-tuned control over specific features of Red Hat Lightspeed to suit the needs of your organization.

If you are an Organization Administrator, you can use the User Access feature under **Identity & Access Management → User Access** in the Hybrid Cloud Console to:

- Control user permissions and organize roles.
- Create groups that include roles and their corresponding permissions.
- Assign users to these groups, allowing them to inherit the permissions associated with their group's roles.

Additional resources

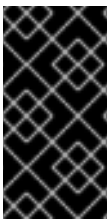
- User Access in the Hybrid Cloud Console
- User Access Configuration Guide for Role-based Access Control (RBAC)

1.6.2. Predefined groups in User Access

The two predefined groups available in User Access, **Default access** and **Default admin access** provide different baseline privilege levels for users in your organization. You can also create custom groups to align permissions with specific personas, job functions, or teams in your organization.

The Default access group

By default, the **Default access** group is assigned many granular predefined roles, so that group members have basic visibility. Because all users in your organization are members of the **Default access** group, they inherit all permissions assigned to that group. The **Default access** group is automatically updated by Red Hat.



IMPORTANT

If your Organization Administrator modifies the **Default access** group, for example, by removing roles to restrict access to specific applications or to use the consolidated roles, the group is automatically renamed to **Custom default access**. Once converted, this group is no longer automatically updated by Red Hat.

The Default admin access group

The **Default admin access** group contains only users who have Organization Administrator permissions. This group is automatically maintained, and users and roles in this group cannot be changed.

The **Default admin access** group includes many (but not all) predefined roles that provide update and delete permissions. The roles in this group usually include **administrator** in their names.

TIP

To view the roles included in the **Default access** or **Default admin access** group, log in to the Hybrid Cloud Console, navigate to **Groups** in User Access, and select the group.

Additional resources

- User Access groups

1.6.3. Predefined roles assigned to groups

Understand how predefined roles in Red Hat Hybrid Cloud Console bundle permissions across multiple Red Hat Lightspeed applications to align with common user personas. Use predefined roles to reduce administrative effort, or create custom roles for more fine-tuned control over specific features.

The predefined roles are a starting point to help you control and manage user permissions. You can then use these roles to create custom roles that are tailored to your specific use cases and organization. For example, you can use the predefined granular roles to create custom roles that provide more fine-tuned control over specific features of Red Hat Lightspeed.

By default, Red Hat provides a set of *consolidated roles* and a set of *granular* roles in the Red Hat Hybrid Cloud Console User Access UI. The consolidated roles significantly reduce the administrative effort required to manage user permissions, while the granular roles provide more fine-tuned control over specific features of Red Hat Lightspeed.

You can use the predefined consolidated and granular roles in User Access simultaneously, but using consolidated roles can significantly reduce the administrative effort.

Select from the predefined consolidated roles library

The Red Hat Hybrid Cloud Console provides three predefined, consolidated User Access roles to help you manage user permissions to Red Hat Lightspeed applications and services that run on registered Red Hat Enterprise Linux systems. These roles help simplify how the Organization Administrator creates groups and permissions for various levels of access to the Red Hat Lightspeed services. If you want to reduce the administrative effort required to manage user permissions and your use case aligns with the permissions included in these roles, select from the consolidated roles library.
The consolidated roles are as follows:

RHEL viewer: The *RHEL viewer* role provides users visibility without the ability to make changes. It allows read-only access to Red Hat Lightspeed. You can view system configurations, compliance reports, inventory data, patch information, vulnerabilities, and overall resource states and activities. The only action permitted with this role is to generate activation keys.

RHEL operator: The *RHEL operator* role allows active management of your Red Hat Lightspeed environment. With this role, you can edit system configurations, inventory details, policies, and notification/integration settings. The RHEL operator role allows many of the RHEL administrator role functions, but it is restricted from editing compliance policies, content source templates, policies, or tasks. In addition, the RHEL operator role cannot execute remediation plans.

RHEL administrator: The *RHEL administrator* role provides comprehensive administrative privileges across your RHEL systems and Red Hat Lightspeed. With this role, you can manage system configurations, inventory, compliance policies, notifications, patch management, remediations, malware detection, and advisor recommendations. The role can also view and modify all vulnerability settings.



IMPORTANT

To use the consolidated roles effectively, you might need to remove the granular RHEL roles from the **Default access** group to prevent permission conflicts. This action automatically changes the name of the predefined **Default access** group to **Custom default access** group, after which, it is no longer automatically updated by Red Hat.

See *Predefined User Access roles* for a list of the roles included in the **Default admin access** group and a reference table that lists most of the predefined groups and roles that are available in the Red Hat Hybrid Cloud Console and the permissions included in each role.

Granular roles

The granular roles are specific roles for individual services that allow for fine-tuned control over specific features of Red Hat Lightspeed, for example, *Inventory Hosts administrator* or *Compliance viewer*. If you want to have more control over specific features of Red Hat Lightspeed and your use case does not align with the permissions included in the consolidated roles, use the granular predefined roles.

TIP

Across the Red Hat Lightspeed product documentation, the **Prerequisites** section for each procedure lists which predefined roles provide the permissions needed to use the features in that procedure. For example, if a procedure requires permissions to create and view remediations, the **Prerequisites** section for that procedure lists the *Remediations user* or other valid role as a recommended predefined role to use for that procedure.

Additional resources

- Preparing the Default access group for consolidated role permissions
- Predefined User Access roles

1.6.4. Check your permissions

Verify your current permissions and the roles or groups assigned to you in the Red Hat Hybrid Cloud Console. Check your permissions to troubleshoot access issues or understand your level of access to Red Hat Lightspeed applications.



NOTE

Only users with the Organization Administrator role can view the permissions of other users in the User Access settings and manage user permissions to Red Hat Lightspeed services. For more information, see the *Configure user permissions* section.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ My User Access**.

2. Optional: If you require additional permissions, use the Red Hat Hybrid Cloud Console Virtual Assistant to ask **"Contact my Organization Administrator"**. The assistant sends an email to the Organization Administrator on your behalf.

Verification

All of the applications that you have permissions to access are listed on this page and are grouped by product, for example, RHEL, OpenShift Container Platform, and Ansible Automation Platform.

You can also filter your permissions by application, for example, by advisor, cost management, inventory, and remediations.

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC)
- Procedures for configuring User Access

1.6.5. Configure user permissions

If you are an Organization Administrator, you can view and manage user permissions for all users in your organization. Control access to Red Hat Lightspeed and other Red Hat Hybrid Cloud Console services through the User Access interface.



IMPORTANT

If you are not an Organization Administrator, you are unable to complete this task. However, you can check your own permissions for different applications by navigating to **Settings → Identity & Access Management→ My User Access**. Contact your Organization Administrator to request more permissions.

Prerequisites

- You have logged in to the Red Hat Hybrid Cloud Console as an Organization Administrator, or you have the required administrator User Access role permissions.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ User Access**.

Verification

From here, you can create and manage:

- **Settings → Identity & Access Management→ User Access → Roles** to determine permissions to Red Hat Lightspeed services and features
- **Settings → Identity & Access Management→ User Access → Groups** to include one or more roles to align with a specific persona, job function, or team in your organization
- **Settings → Identity & Access Management→ User Access → Users** and their assignment to groups to inherit permissions from the roles assigned to those groups

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC)
- Procedures for configuring User Access
- Simplifying access management on Hybrid Cloud Console for Red Hat Lightspeed with predefined roles (Red Hat Knowledgebase)

1.6.6. User Access roles for permissions to create and execute remediation plans

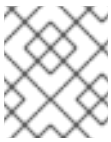
Understand the predefined roles that control permissions to view, create, and execute remediation plans on your RHEL systems. Use these role definitions to assign appropriate permissions to users based on their responsibilities.

To remediate issues on your RHEL systems, become familiar with the roles that provide the required access permissions for creating, managing, and executing remediation plans in Red Hat Lightspeed.

The following User Access roles provide standard or enhanced access to remediation features in Red Hat Lightspeed:

Table 1.1. Permissions provided by the User Access roles

User Access role	Grants permissions to ...	Included in the Default access group
Remediations administrator	• Do everything that a **Remediations user** can do. • Execute remediation plans on connected remote host systems from within Red Hat Lightspeed.	
Remediations user	• View existing remediation plans. • Create a remediation plan. • Delete a remediation plan.	X
RHEL administrator	• Do everything that a **RHEL operator** can do. • Administer RHEL system configs, inventory, compliance, notifications, patch management, execute remediation plans, malware detection, and advisor. • View and modify vulnerability settings.	

User Access role	Grants permissions to ...	Included in the Default access group
RHEL operator	• Do everything that a **RHEL viewer** can do. • Edit system configs, inventory, policies, notifications, and integrations. • View compliance reports, patch info, malware detections, and recommendations. • Create remediation plans, manage stale data, and change vulnerability settings.  **NOTE** The RHEL operator role is restricted from editing compliance policies, content source templates, policies, or tasks. Also, the RHEL operator role cannot execute remediation plans.	
RHEL viewer	• Read all available data across Red Hat Lightspeed services and features. ◦ View system configs, compliance reports, inventory data, patch info, vulnerabilities, and more to observe the state of resources or activities.  **NOTE** Cannot perform actions other than generating activation keys.	

For more information about user access and permissions, see User Access Configuration Guide for Role-based Access Control (RBAC).

CHAPTER 2. ENABLE HOST COMMUNICATION WITH RED HAT LIGHTSPEED

Before you can execute the playbooks in your remediation plans on remote systems from Red Hat Lightspeed, your systems must be able to communicate with Red Hat Lightspeed.

- For Red Hat Enterprise Linux systems that are *not managed by Red Hat Satellite* , you should complete the following procedure to enable the rhc client on those systems.
- For systems that *are managed* by Satellite, you configure Cloud Connector on the host servers for those systems.

2.1. ENABLE THE RHC CLIENT ON SYSTEMS DIRECTLY MANAGED BY RED HAT LIGHTSPEED

To execute Ansible Playbooks from Red Hat Lightspeed, enable the rhc client on the systems in your infrastructure. The **rhc connect** command does this by registering (RHEL8.6 and later, and 9.0 and later) systems with Red Hat Subscription Manager and Red Hat Lightspeed, and enabling remote host configuration (rhc) features in Red Hat Lightspeed.

Prerequisites

- Sudo access on the Red Hat Enterprise Linux host system

Procedure

1. To connect rhc on RHEL 8.5 systems, use the following commands:

```
# subscription-manager repos --enable ansible-2.9-for-rhel-8-x86_64-rpms
# dnf -y install ansible rhc-worker-playbook-0.1.5-3.el8_4
# rhc connect
```



NOTE

Remote host configuration on RHEL 8.5 has dependencies on **ansible** and **rhc-worker-playbook**. To install the dependencies, you must first register with Subscription Manager.

2. To connect rhc on RHEL8.6 and later systems, use the following commands:

```
# dnf -y update rhc
# dnf -y install rhc-worker-playbook
# rhc connect
```

3. To connect rhc on RHEL9.0 and later systems, use the following commands:

```
# dnf -y install rhc rhc-worker-playbook
# rhc connect
```

Additional resources

- Red Hat Hybrid Cloud Console > Red Hat Lightspeed > Inventory > System Configuration > Remote Host Configuration (RHC)
- Registering RHEL systems and configuring client tools with Red Hat Lightspeed

2.2. ENABLE REMOTE HOST CONFIGURATION MANAGER IN THE RED HAT LIGHTSPEED UI

To allow users to execute a remediation plan on a remote system from Red Hat Lightspeed, you must configure the Remote Host Configuration Manager settings in the Red Hat Lightspeed UI.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with at least one of the following roles:
  - **RHC Administrator**
  - **Inventory Hosts administrator**
  - **RHEL administrator**
    Contact your Organization Administrator to obtain the required permissions.

Procedure

1. Navigate to Remote Host Configuration Manager.
2. Under **Permission**, set the option **Allow permitted Red Hat Lightspeed users to execute remediation playbooks on rhc-connected systems** to **Enabled**.
3. Click **Save changes**.

Verification

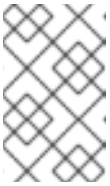
A confirmation box with the message **Changes saved** displays at the top right corner of the UI.

2.3. CHECK THE CONNECTION STATUS OF HOST SYSTEMS IN THE RED HAT LIGHTSPEED UI

After you set up **rhc connect** and enable host communication, verify that the host system is connected in the Red Hat Lightspeed UI.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with at least one of the following roles:
  - **Remote Host Configuration (RHC) Administrator**
  - **Inventory Hosts administrator**
  - **RHEL administrator**



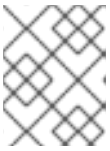
NOTE

If you need to execute a remediation plan on remote hosts from Red Hat Lightspeed, you also need at least the **Remediations administrator** or **RHEL administrator** predefined User Access role.

Contact your Organization Administrator to obtain the required permissions.

Procedure

1. In the Red Hat Hybrid Cloud Console, navigate to Automation Toolkit > Remediation Plans.
2. On the **Remediation Plans** page, choose a remediation plan that includes the system you are interested in checking. The **General** tab opens.
3. Open the **Systems** tab, and locate the system in the table.
4. Check the **Connection Status** value for the system.



NOTE

If the **Connection Status** value is **Unknown**, this means that you do not have the correct User Access permissions to view the status.

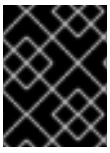
Additional resources

- Enabling the **rhc** client on systems directly managed by Red Hat Lightspeed

2.4. ENABLE CLOUD CONNECTOR FOR CONTENT HOSTS MANAGED BY SATELLITE

You can remediate issues on Satellite-managed content hosts remotely from the Red Hat Lightspeed UI in the Red Hat Hybrid Cloud Console. Remote remediation from Red Hat Lightspeed requires that you first configure the Cloud Connector plugin on the Satellite Server and must have the following:

- Satellite must be version 6.9 or later.
- You have root access to the Satellite server.
- The content hosts that are managed by the Satellite should have the insights-client installed and turned on. For insights-client installation and enablement procedures, see *Additional resources*.
- Import a subscription manifest into Satellite.
- Register your hosts to Satellite using an activation key to attach Red Hat subscriptions.



IMPORTANT

If you want to manage and execute host remediations entirely from the Satellite web UI, you do not need to enable the Cloud Connector plugin.

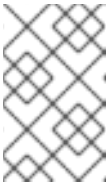
2.4.1. Configure Cloud Connector and upload your Satellite Server content host inventory to Red Hat Lightspeed

Before you can run the playbooks in your remediation plans remotely from within Red Hat Lightspeed, you must install and configure the Cloud Connector plugin on the Satellite Server.

Complete the following tasks to install, configure, and verify the configuration of Cloud Connector.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with at least one of the following roles:
  - **RHC Administrator**
  - **Inventory Hosts administrator**
  - **RHEL administrator**



NOTE

If you need to execute a remediation plan on remote hosts from Red Hat Lightspeed, you also need at least the **Remediations administrator** or **RHEL administrator** predefined User Access role.

Contact your Organization Administrator to obtain the required permissions.

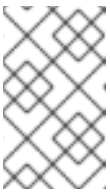
Procedure

1. On Satellite Server, enable the **remote-execution** plugin by entering one of the following commands, based on your version of Satellite Server.
   a. On Satellite Server 6.12 and later.

```
[root]# satellite-installer --foreman-proxy-plugin-remote-execution-script-install-key true
```

   b. **On Satellite Server 6.9 - 6.11**

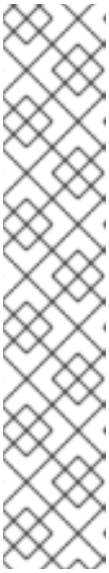
```
[root]# satellite-installer --foreman-proxy-plugin-remote-execution-ssh-install-key true
```



NOTE

Configuring Cloud Connector requires that the Satellite perform a remote execution on itself. This is why the first step is to enable the **remote-execution** script or plugin.

2. In the Satellite web UI, navigate to **Configure > Red Hat Cloud > Inventory Upload**Verify that the **Automatic Inventory Upload** switch is turned **ON**, which is the default setting.
3. **Optional:** Toggle the **Obfuscate host names** switch to the **ON** position to hide hostnames that Satellite reports to the Hybrid Cloud Console.



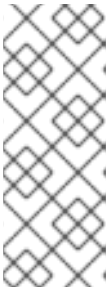
NOTE

The **Obfuscate host names** setting only affects **rh_cloud** reports. If you want to obfuscate hostnames, IP addresses, or Media Access Control (MAC) addresses, configure obfuscation in the insights-client configuration. Satellite observes the Red Host Lightspeed obfuscation configuration settings. For more information, see the following resources:

- *Obfuscating hostnames*
- *Obfuscating IP addresses* .
- *Obfuscating MAC addresses* .

Automatic inventory upload and **Obfuscate host names** are global settings. They affect content hosts that belong to all organizations.

4. Click **Configure Cloud Connector**. A Notice dialog box warns you that this action also enables auto reports upload.
5. Click **Confirm**, and then wait for the task to finish. This should take about one minute.
6. Go to **Monitor > Jobs > Configure Cloud Connector** to see the job.



NOTE

- Eventually, you see the Satellite in Red Hat Hybrid Cloud Console > the **Settings** icon (⚙) > Integrations, on the **Red Hat** tab. Allow up to one hour after the job is visible in the Satellite web console.
- The bottom of the **Inventory Uploads** page shows the name of your organization; hovering over it turns the area gray.

7. Click **Generate and upload report** to generate a data payload from each of the content hosts that have insights-client running, and upload your host inventory to Red Hat Lightspeed.
8. Repeat the previous step for each organization for which you want to upload a content host inventory.
9. Under **Configure** > **Red Hat Cloud (after Sat 6.11)**> **Red Hat Lightspeed**, set **Auto sync** for the organization by using the toggle in the upper right corner of the screen.

Verification

To verify that the upload was successful, log in to Red Hat Hybrid Cloud Console > Red Hat Enterprise Linux > Red Hat Lightspeed > Inventory and search for the **satellite_id** tag for your content hosts.

Optional: Click **Sync all inventory status** and wait for the task to finish. It shows you the number of content hosts recognized by Red Hat Lightspeed inventory.

2.4.2. Additional resources

- Importing a Subscription Manifest into Satellite Server

CHAPTER 3. USE TOKEN-BASED SERVICE ACCOUNTS WITH REMEDIATION PLANS

If you use Ansible Automation Platform to view, download, and execute remediation plans, you can configure a token-based service account for use with both Red Hat Lightspeed and Ansible Automation Platform. This configuration provides a more secure, scalable, and automation-friendly integration.

In addition, you can view the remediation plans associated with the service account in both Red Hat Lightspeed and Ansible Automation Platform.

3.1. SERVICE ACCOUNTS FOR REMEDIATIONS

After you set up the service account and configure User Access in Red Hat Lightspeed, the service account can securely access all remediation plans connected to that service account. You can also view, download, and execute playbooks for remediations from within Ansible Automation Platform.



NOTE

Token-based service accounts replace Basic Authentication, which is no longer supported for connecting to the Red Hat Hybrid Cloud Console and Red Hat Lightspeed APIs.

Additional resources

- Support for token-based authentication via Service Account for Red Hat Lightspeed (Insights) in Ansible Automation Platform
- Integrate with Red Hat Lightspeed (formerly Insights)
- Setting up Red Hat Lightspeed Remediations

3.2. CONFIGURE THE SERVICE ACCOUNT

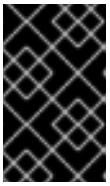
You can create a new token-based service account in the Red Hat Hybrid Cloud Console to integrate with both Red Hat Lightspeed and Ansible Automation Platform, or you can select an existing service account.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as an Organization Administrator.

Procedure

1. In the Red Hat Hybrid Cloud Console, navigate to Red Hat Hybrid Cloud Console > the **Settings** icon (⚙) > Service Accounts.
2. Create a token-based service account, or select an existing service account. For more information about creating service accounts, see Creating a service account .



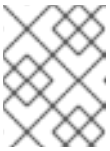
IMPORTANT

If you create a new service account, ensure that you save the Client ID and Client secret to a safe location. If you select an existing service account, ensure that you have access to the Client ID and Client secret.

3. Create a User Access group to associate to the service account, or assign the service account to an existing group that has the required permissions. For more information about creating User Access groups, see Managing group access with roles and members.

4. Assign the following permissions to the group, if the group does not already have them. For more information about how to add roles and permissions to a User Access group, see Adding a role to a group.

   - **inventory:hosts:read** (included in the Inventory Hosts viewer role)
   - **patch:*:read** (included in the Patch viewer role)
   - **remediations:remediation:read** and **playbook-dispatcher:run:read** (included in the Remediations User role)

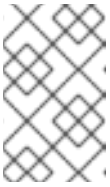


NOTE

You can also grant the *RHEL viewer* role to the service account in the User Access group. The *RHEL viewer* role includes the following permissions:

   - **inventory:hosts:read**
   - **remediations:remediation:read**

For more information about assigning a service account to a User Access group, see Adding service accounts to a User Access group.



NOTE

If your organization uses Workspaces, ensure that your User Access group has the necessary permissions for full visibility into your inventory in AAP. For more information about workspaces, see Workspaces.

Additional resources

- Creating and managing a service account
- Support for token-based authentication via Service Account for Red Hat Lightspeed (Insights) in Ansible Automation Platform
- Managing group access with roles and members
- Adding service accounts to a User Access group
- Workspaces

3.3. CONFIGURE CREDENTIALS IN THE ANSIBLE AUTOMATION PLATFORM

After you configure the service account and User Access in the Red Hat Hybrid Cloud Console, you can create credentials in Ansible Automation Platform.

Prerequisites

- The Client ID and Client secret for the service account, which you obtained when you created or selected a service account in the Red Hat Hybrid Cloud Console.
- Access to the Ansible Automation Platform (AAP) interface.

Procedure

1. Create a new credential in AAP. The **Create credential** screen displays. For more information about how to create and configure credentials in AAP, see Creating Red Hat Insights credentials.
2. In the **Credential Type** drop-down menu, select **Insights** as the credential type.
3. Paste the Client ID and Client secret for the service account into the corresponding fields in the **Type Details** section.
4. Click **Create credential**.

Additional resources

- Creating and managing a service account
- Creating Red Hat Insights credentials

3.4. VIEW THE REMEDIATION PLANS ASSOCIATED WITH THE SERVICE ACCOUNT

After you set up the service account in both Red Hat Lightspeed and Ansible Automation Platform, you can view the remediation plans associated with the service account in Red Hat Lightspeed.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as an Organization Administrator.

Procedure

- In the Red Hat Hybrid Cloud Console, navigate to Automation Toolkit > Remediation Plans. The remediation plans associated with the service account display in the Remediation Plan table.

Additional resources

- Viewing remediation plans

CHAPTER 4. CREATE A REMEDIATION PLAN IN RED HAT LIGHTSPEED

To fix issues that Red Hat Lightspeed identifies for a system or group of RHEL systems in your organization, create a remediation plan.

4.1. REMEDIATION PLANS IN RED HAT LIGHTSPEED

You can create a remediation plan to fix one or more issues identified by Red Hat Lightspeed for a system or group of RHEL systems in your organization.

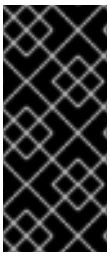
To create a remediation plan in Red Hat Lightspeed, you need to do the following:

1. Find an issue to resolve
2. Review the recommended remediation steps
3. Select the systems to remediate

You can create a remediation plan to address recommendations and issues found by the following services of Red Hat Lightspeed:

- advisor
- compliance
- vulnerability
- patch

The workflow to create a remediation plan is similar for all services in Red Hat Lightspeed that support remediations.



IMPORTANT

For Red Hat recommended solutions that can only be resolved by applying a manual remediation, the advisor recommendation displays a **Remediation type** value of **Manual**.

You can create a remediation plan for any Red Hat Lightspeed recommendations or remedial actions that have a **Remediation type** value of **Playbook**.

When you create a remediation plan, Red Hat Lightspeed generates an Ansible Playbook that enables the required remediation actions and reboot instructions on the selected host systems.

Additional resources

- Remediations overview

4.2. CREATE A REMEDIATION PLAN FOR AN ADVISOR SERVICE RECOMMENDATION

You can create a remediation plan to address advisor service recommendations.

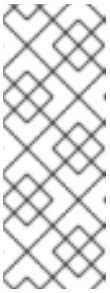
The advisor service assesses and monitors the health of your Red Hat Lightspeed infrastructure and provides recommendations to address availability, stability, performance, and security issues. Red Hat Lightspeed detects the systems in your infrastructure that are impacted and provides a set of recommended actions that can help you prioritize and plan how to remediate your systems.

For more information about the Red Hat Lightspeed advisor service, see Assessing RHEL configuration issues using the Red Hat Lightspeed advisor service.

To create a remediation plan that addresses a recommendation generated by the Red Hat Lightspeed advisor service, complete the following procedure.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with at least one of the following roles:
  - **Remediations user**
  - **RHEL viewer**



NOTE

By default, all Red Hat Lightspeed users have permissions to create remediation plans. However, you need at least the **Remediations administrator** role or the **RHEL administrator** predefined User Access role to execute a remediation plan on remote hosts from Red Hat Lightspeed. For more information, see the Required permissions for remediation plan execution.

Procedure

1. Choose a Red Hat Lightspeed advisor service recommendation to remediate:
   a. Navigate to Operations > Advisor > Recommendations.
   b. Review the **Recommendations** table to see which recommendations are applicable for your systems and whether they have a playbook already created.
   c. Use the search and filtering function in the table to sort the items by **Remediation type**.



IMPORTANT

Look for recommendations that have a **Remediation type** of **Playbook**. You cannot create a remediation plan if the **Remediation type** is set to **Manual**.

   d. Click the recommendation name. The full details of the recommendation are displayed, and a list of impacted systems is displayed on the lower part of the page.
2. Select which systems to include in the remediation plan:
   a. Scroll to view all of the registered RHEL systems that are impacted by the recommendation.
   b. Find the systems to include. If needed, use the search and filter functions in the table. For example, you can use the filtering options to list the affected systems by version.



IMPORTANT

To create a remediation plan for a group of systems, all systems in a group must be running the same RHEL major and minor versions. This ensures that the Red Hat Lightspeed-generated playbook resolution is compatible.

c. Select at least one system to include in a remediation plan by clicking the checkbox to the left of the system ID.

3. Create the plan:

   a. Click **Plan remediation**.

   b. Enter a new plan name in the **Plan name** field or start typing to select an existing plan.

   c. Click **Create plan** for a new plan or **Update plan** for an existing plan.

Verification

1. Navigate to Automation Toolkit > Remediation Plans.
2. Search for your remediation plan in the table and click its name to open the plan.

4.3. CREATE A REMEDIATION PLAN FOR A CVE VULNERABILITY ON RHEL SYSTEMS

You can create a remediation plan in the Red Hat Lightspeed vulnerability service. When you create a remediation plan, Red Hat Lightspeed uses Ansible Playbooks to remediate or mitigate CVE vulnerabilities on your systems and apply any required patches.

The Red Hat Lightspeed advisor service analyzes and detects which systems in your organization are affected by known problems.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with at least one of the following roles:
  - **Remediations user**
  - **RHEL viewer**



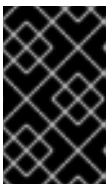
NOTE

By default, all Red Hat Lightspeed users have permissions to create remediation plans. However, you need at least the **Remediations administrator** role or the **RHEL administrator** predefined User Access role to execute a remediation plan on remote hosts from Red Hat Lightspeed. For more information, see the Required permissions for remediation plan execution.

Procedure

1. Navigate to the Security > Vulnerability > CVEs page.

2. Set the filters as needed and select a CVE.
3. Scroll down to view all of the affected systems. Use the filtering options to list the affected systems by version.
4. Select systems to include in a remediation plan by clicking the box to the left of the system ID.



IMPORTANT

To create a remediation plan for a group of systems, all systems in a group must be running the same RHEL major and minor versions. This ensures that the Red Hat Lightspeed-generated playbook resolution is compatible.

5. Click **Plan remediation**.
6. Choose whether to add the remediations to an existing or new remediation plan, and then do one of the following actions, and then click **Next**:
   - Click **Add to existing playbook**, and then select a remediation plan from the list presented.
   - Click **Create new playbook**, and enter a name for the playbook.
7. Review the systems to include in the remediation plan, then click **Next**.



NOTE

Only affected systems can be selected and included in a remediation plan.

8. Review the information under the remediation review summary.
   a. If a reboot is required to fix the issue or risk, all systems in the remediation plan are automatically rebooted. If you prefer to reboot manually after the plan has been executed, toggle the **Auto-reboot** button accordingly.
   b. Click **Submit**.

Verification

1. Navigate to Automation Toolkit > Remediation Plans.
2. Search for your remediation plan. You should see the plan that you created showing in the list.

4.4. REMEDIATE CVES WITH SECURITY RULES THAT HAVE RECOMMENDED AND ALTERNATE RESOLUTION OPTIONS

Most CVEs in Red Hat Lightspeed have one remediation option for you to use to resolve an issue. Remediating a CVE with security rules might include more than one resolution from which to choose.

For example, you might have a recommended action to take, and one or more alternate resolutions. The workflow to create remediation plans for CVEs that have one or more resolution options is similar to the remediation steps in the advisor service.

For more information about security rules, see Security rules and Filtering lists of systems exposed to security rules in *Assessing and monitoring security vulnerabilities on RHEL systems*

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with at least one of the following roles:
  - **Remediations user**
  - **RHEL viewer**

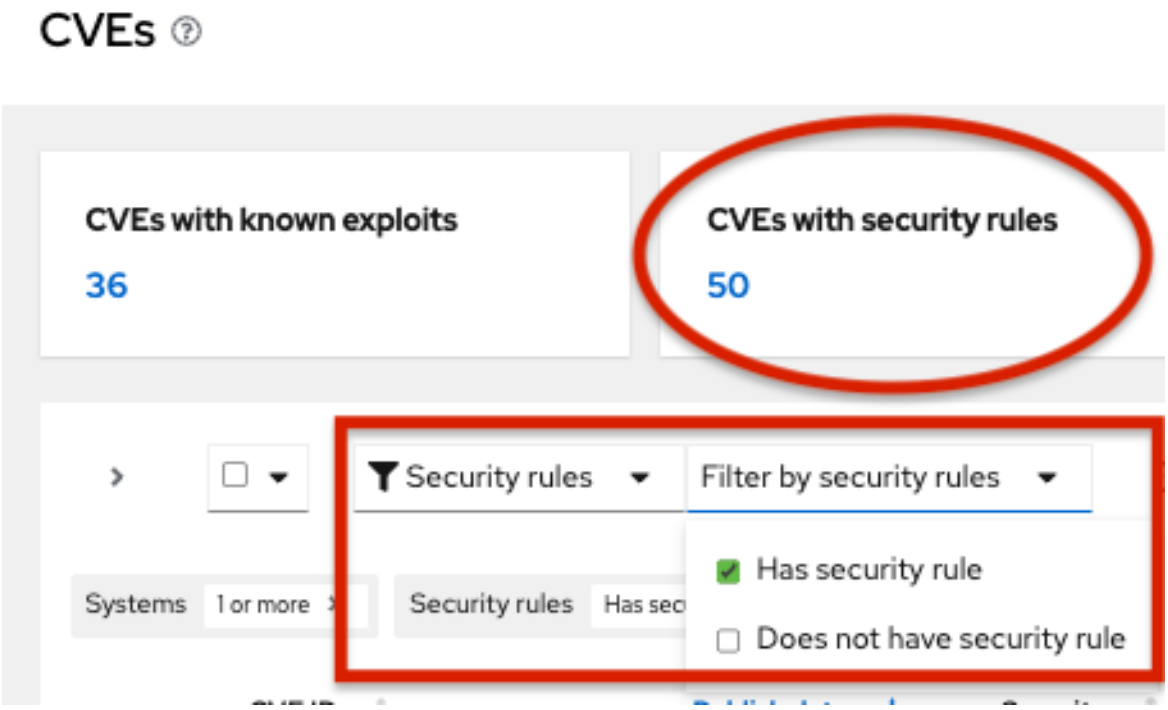


NOTE

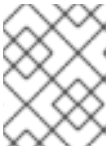
By default, all Red Hat Lightspeed users have permissions to create remediation plans. However, you need at least the **Remediations administrator** role or the **RHEL administrator** predefined User Access role to execute a remediation plan on remote hosts from Red Hat Lightspeed. For more information, see the Required permissions for remediation plan execution.

Procedure

1. Navigate to **Security > Vulnerability > CVEs**
2. Set filters if needed (for example, filter to see **CVEs with security rules** to focus on issues that have elevated risk associated with them). Or, click the CVEs with security rules tile on the dashbar.



3. Click a CVE in the list.
4. Scroll to view affected systems and select the systems to include in a remediation plan. Click the box to the left of each system ID on the **Review systems** page. Selecting at least one system activates the **Plan remediation** button.



NOTE

Recommended: Include systems of the same RHEL major or minor version by filtering the list of affected systems.

5. Click **Plan remediation**.
6. Decide whether to add the selected remediations to an existing or new remediation plan by taking one of the following actions:
   - Click **Add to existing playbook** and select the required playbook from the dropdown list.
   - Click **Create new playbook**, and add a playbook name.
7. Click **Next**. The systems impacted by the CVE are listed. Review the systems to include in the playbook and clear the checkbox beside any systems that you do not want to include.



NOTE

Only impacted systems can be selected and included in a remediation plan.

8. Click **Next** to see the **Review and edit actions** page, which shows you options to remediate the CVE. The number of items to remediate can vary. You also see additional information (that you can expand and collapse) about the CVE, such as:
   - **Action:** Shows the CVE ID.
   - **Resolution:** Displays the recommended resolution for the CVE and also confirms whether you have alternate resolution options.
   - **Reboot required:** Confirms whether you must reboot your systems.
   - **Systems:** Confirms the number of systems you are remediating.
9. On the **Review and edit actions** page, choose one of two options to finish creating your remediation plan and to generate the Ansible Playbook:
   - **Option 1:** To review all of the recommended and alternative remediation options available (and choose one of those options):
     a. Select **Review and/or change the resolution steps for this 1 action** or similar based on your actual options.
     b. Click **Next**.
     c. On the **Choose action: <CVE information>** page, click a tile to select your preferred remediation option. The bottom edge of the tile highlights when selected. The recommended solution is highlighted by default.
     d. Click **Next**.
   - **Option 2:** To accept all recommended remediations:
     a. Choose **Accept all recommended resolution steps for all actions**
     b. Click **Next**.

10. On the final **Remediation review** pane, review the summary of your remediation plan and use the back button to make changes to the actions or resolution options if required. When you are satisfied with the plan, click **Submit**.



NOTE

If a reboot is required to fix the issue or risk, all systems in the remediation plan are automatically rebooted. If you prefer to reboot manually after the plan has been executed, toggle the **Auto-reboot** button accordingly.

Verification

A notification confirming the total number of remediation actions and other information about your remediation plan is displayed.

Next steps

To view your remediation plan:

1. Navigate to **Automation Toolkit > Remediation Plans**
2. Search for your remediation plan.
3. To execute your remediation plan and run the generated Ansible Playbook on the affected systems, see Executing remediation playbooks from the Red Hat Lightspeed UI .

CHAPTER 5. VIEW AND MANAGE YOUR REMEDIATION PLANS

Red Hat Lightspeed provides a central location under Automation Toolkit > Remediation Plans to help you find, view the full details of, and manage the remediation plans that have been created for your organization. You can also download, delete, or modify your remediation plans from the main **Remediation Plans** page.



IMPORTANT

You cannot create a remediation plan from Automation Toolkit > Remediation Plans. To create a new remediation plan, you must use the Red Hat Lightspeed service that generated the recommendation and its underlying Ansible Playbook, such as the advisor, compliance, or vulnerability service.

5.1. VIEW REMEDIATION PLANS

The **Remediation Plan** view provides a comprehensive overview of all of the remediation plans that you created for your organization in Red Hat Lightspeed.

5.1.1. About Remediation plans table view

You can access a table listing of all your remediation plans by navigating to the **Automation Toolkit** page and selecting **Remediation Plans**.

The table view in Remediation Plans provides a quick overview of all remediation plans, including their status and last execution date.

You can search, sort, and filter the list of all of the remediation plans for your organization. For example, filter the table to show only specific remediation plans based on status.

By default, the columns in the table contain the following information:

Name
: The name of the remediation plan.

Last Executed
: The date and time when the remediation plan was last executed.

Execution Status
: The latest execution status of the remediation plan, for example, successful or failed. N/A indicates that the remediation plan has not been executed yet.

Actions
: The number of actions that run when the remediation plan is executed.

Systems
: The number of systems that the remediation plan is selected to run on.

Expiration
: Time remaining until the remediation plan is automatically deleted due to inactivity.

Created
: The date and time when the remediation plan was created.

Last Modified
: The date and time when the remediation plan was last modified.

5.1.2. Customize the remediation plan table

You can customize the **Remediation Plan** table to suit your needs. You can add or remove columns and sort the table by any column.

Procedure

1. Navigate to Automation Toolkit > Remediation Plans.
2. Click the options icon (⋮) beside the **Download** button and select **Manage columns** to select which columns to display in the table.
3. Click **Reset to Default** to restore the default view.

5.1.3. Remediation plan details

When you select and click a remediation plan in the table view, all of the available details about the plan are displayed across three tabs.

General

Contains three cards that display the remediation plan status and summary details:

- **Overview**: Displays the plan name, total number of actions, total number of systems included, and auto-reboot configuration. It also contains a **Learn more** link to the support documentation.
- **Activity**: Displays latest execution status, date created, expiration, last modified date, and last execution date. The expiration field shows the relative time remaining until the plan is automatically deleted and includes a help icon explaining the retention policy. If a plan is nearing deletion, an expiration warning alert, for example, "Expires in 6 days", displays here.
- **Execution readiness summary**: Displays the execution readiness checklist results. It also contains a **Learn more** link to a quick start guide.

Planned remediations

Contains information about remediation plan execution limits and two tabs that display the planned remediation details:

- **Remediation plan execution limits**: Execution of a remediation plan must be within the limit of 1000 action points or 100 systems.
- **Actions**: Displays a table listing the actions included in the plan.
- **Systems**: Displays a table listing the systems included in the plan.

Execution history

Displays a timeline of all executions of the remediation plan, including the dates, status, history, and links to the logs of a plan execution for each included system. From here, you can see any errors that occurred during execution.

You can also download and view the associated playbook by clicking on **Download**.

5.2. DOWNLOAD REMEDIATION PLANS

You can download the generated playbooks for each remediation plan in your organization. When you download a remediation plan, the YAML file for the underlying Ansible Playbook is saved to the preferred download directory on your local browser client.

You might need to download a remediation plan for the following reasons:

- To execute a remediation plan on your host systems by using the external Ansible Automation Platform (AAP) workflow for your organization instead of from within the Red Hat Lightspeed application.
- To view the specific plays of a remediation plan.
- To troubleshoot issues with executing a remediation plan.

Procedure

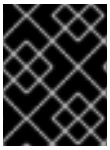
1. In the Red Hat Hybrid Cloud Console, navigate to Automation Toolkit > Remediation Plans.
2. Select the checkbox next to the name of the remediation plan you want to download.
3. Complete one of the following steps:
   - To download a single remediation plan, go to the end of the row, click the options icon (⋮) and select **Download**.
   - To download multiple remediation plans in bulk, click the **Download** button at the top of the table.

Verification

A message is displayed to confirm that the download was successful. A YAML file for each of the selected remediation plans is downloaded to your local drive.

5.3. DELETE REMEDIATION PLANS

You can permanently delete remediation plans that are no longer needed.



IMPORTANT

You cannot recover a deleted remediation plan. Also, you cannot archive and restore a remediation plan from the Red Hat Lightspeed UI.

Procedure

1. In the Red Hat Hybrid Cloud Console, navigate to Automation Toolkit > Remediation Plans.
2. In the table, find the plan you want to delete.
3. At the end of the row, click the **More options** icon, and then click **Delete**.
4. When prompted, click **Delete** to confirm the permanent removal of the plan.

Verification

A message is displayed to confirm the successful deletion of the selected remediation plan.

5.4. RENAME A REMEDIATION PLAN

You can rename an existing remediation plan in your organization.

Procedure

1. In the Red Hat Hybrid Cloud Console, navigate to Automation Toolkit > Remediation Plans.
2. Select the checkbox next to the name of the remediation plan you want to rename.
3. At the end of the row, click the options icon (⋮) and select **Rename**.
4. When prompted, enter a unique title for the plan name and click **Rename**.

Verification

- A message is displayed to confirm the successful renaming of the selected remediation plan.

5.5. CONFIGURE THE ACTIONS IN A REMEDIATION PLAN

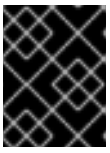
You can modify or remove an action or system from the plan, especially if the plan contains many items.

5.5.1. Add a new recommendation action to an existing remediation plan

If the Red Hat advisor or another service recommends a new remedial action for one or more systems in your organization, you can add that recommendation to an existing remediation plan.

Procedure

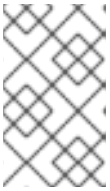
1. Navigate to Operations > Advisor > Recommendations.
2. In the table, use the search and filtering options to find the recommendation you want to add, and click the recommendation name.



IMPORTANT

The recommendation must have a **Remediation type** of **Playbook**. You cannot create a remediation plan if the **Remediation type** is **Manual**.

3. Select at least one impacted system to remediate, and click **Plan remediation**.
4. Select **Add to existing playbook**, and then select the remediation plan name from the list presented. Click **Next**.
5. Review the systems in the plan and clear the checkbox next to any systems that you do not want to include. Click **Next**.
6. Review the remedial actions of your plan and adjust them if necessary. When you are satisfied, click **Next**.
7. On the final **Remediation review** pane, review the summary of your remediation plan, and click **Back** to make changes, if needed.



NOTE

Auto-reboot is enabled if any of the recommended actions to remediate require a system reboot to take effect. If you prefer to reboot manually after the plan has been executed, toggle **Auto-reboot** accordingly.

8. Click **Submit**.

Verification

1. Navigate to Automation Toolkit > Remediation Plans.
2. Search for the remediation plan that you created and click the name to open the details.
3. Click **Actions** to view the remedial actions included in the plan. The recommendations that you added are listed.

5.5.2. Remove a remedial action from a remediation plan

You can remove a remedial action from a remediation plan.

Procedure

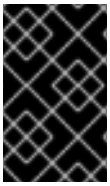
1. In the Red Hat Hybrid Cloud Console, navigate to Automation Toolkit > Remediation Plans. The remediation plans for your organization are displayed.
2. Find the remediation plan that you want to change and click the name of the remediation plan to display the details of the plan.
3. Navigate to the **Actions** tab.
4. In the table, do one of the following:
   - Select the checkbox next to the actions that you want to remove, and click **Remove** at the top of the table.
   - Find a single action to remove, go to the end of the corresponding row, and click **More options**.
5. When prompted, click **Remove**.

Verification

- A message is displayed to confirm that the action was successfully removed from the remediation plan.

5.5.3. Add systems to an existing remediation plan

You can include more systems in an existing remediation plan.



IMPORTANT

Ensure that all systems in the remediation plan are running the same RHEL major and minor versions to ensure that the resolution applied by the Red Hat Lightspeed-generated playbook is compatible.

Procedure

1. Navigate to Operations > Advisor > Recommendations.
2. Use the search and filtering options to find the recommendation in the existing plan and click the recommendation name.
3. Select the systems you want to add to the plan, and then click **Plan remediation**.
4. Select the option to **Add to existing playbook**, and then select the existing remediation plan name from the list presented. Click **Next**.
5. Review the systems in the plan, and if applicable, clear the checkbox next to any systems that you do not want to include. Click **Next**.
6. Review the remedial actions of your plan and adjust if necessary, as outlined in the section titled *Creating a remediation plan to remediate an advisor service recommendation* . Click **Next**.
7. On the final **Remediation review** pane, review the summary of your remediation plan and click **Back** to make changes if required.



NOTE

The **Auto-reboot** button is enabled if any of the recommended actions to remediate require a system reboot to take effect. If you prefer to reboot manually after the plan has been executed, toggle the **Auto-reboot** button accordingly.

8. Click **Submit**.

Verification

1. Navigate to Automation Toolkit > Remediation Plans.
2. Search for your remediation plan in the table, and click the name to open the remediation plan.
3. Click **Systems** to view a list of the systems included in the plan. The systems you added should be listed.

5.5.4. Remove a system from a remediation plan

You can remove a system from a remediation plan.

Procedure

1. In the Red Hat Hybrid Cloud Console, navigate to Automation Toolkit > Remediation Plans.

2. Find the remediation plan to remove a system from and click the name to open the details of the remediation plan.
3. Navigate to the **Planned remediations** tab and open the **Systems** tab.
4. In the table, complete one of the following steps:
   - Select the systems that you want to remove, then click **Remove** above the table.
   - Find a single system to remove, go to the end of the corresponding row, click **More options**.
5. When prompted, click **Remove**.

Verification

- A message is displayed to confirm that the system was successfully removed from the remediation plan.

5.6. REMEDIATION PLAN RETENTION POLICY OVERVIEW

Inactive remediation plans are automatically deleted after 4 months to prevent accumulation of stale plans. The expiration timer resets when plans are modified or executed. Organization Administrators can configure retention settings.

5.6.1. Remediation plan retention policy

To help you manage your active work, reduce storage requirements, and improve database performance, Red Hat Lightspeed includes an automated retention policy for remediation plans. This policy prevents the accumulation of stale and outdated plans.

By default, inactive remediation plans are automatically deleted after 4 months (120 days) of inactivity. The expiration timer resets whenever you modify or run the remediation plan.

By default, an expiration warning displays in the Red Hat Lightspeed UI 30 days before a plan is scheduled for deletion.



NOTE

System email notifications regarding impending plan deletions are out of scope for this feature. You must review the Red Hat Lightspeed user interface to monitor expiration warnings.

If you have Organization Administrator privileges, you can edit the default retention period and warning thresholds to meet your organizational compliance requirements.

5.6.2. Configure the remediation plan retention policy

By default, remediation plans are deleted after 4 months of inactivity. You can modify the default retention period and the expiration warning threshold to suit your organization's needs.

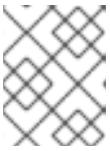
Changes to the retention policy affect all user accounts and remediation plans across your organization.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user with **Organization Administrator** privileges.

Procedure

1. Navigate to Automation Toolkit > Remediation Plans.
2. Click the options icon (⋮) beside the **Launch Quick Start** button and select **Edit retention policy**.
3. In the **Remediation plan retention policy** modal, click the **Retention period** drop-down menu and select the duration that an inactive remediation plan is kept before being automatically deleted. Options range from 14 days to 6 months.
4. Click the **Expiration warning** drop-down menu and select the duration to display a warning before an inactive plan is automatically deleted. Options range from 3 days to 5 months.



NOTE

The expiration warning period cannot be longer than the configured retention period.

5. Click **Save**.

Verification

- A confirmation message opens: **Remediation plan retention policy updated**

5.7. MONITOR REMEDIATION STATUS

You can view the latest execution status and activity for each remediation plan that you execute from the Red Hat Lightspeed application. You can also view the execution logs for the remediation plan and see the results of the Ansible Playbook execution.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with at least one of the following roles:
  - **Remediations user**
  - **RHEL viewer**



NOTE

By default, all Red Hat Lightspeed users have permissions to create remediation plans. However, you need at least the **Remediations administrator** role or the **RHEL administrator** predefined User Access role to execute a remediation plan on remote hosts from Red Hat Lightspeed. For more information, see the Required permissions for remediation plan execution.

Procedure

1. Navigate to Automation Toolkit > Remediation Plans.
2. In the table, find the plan that you want to monitor.
3. Click the name of the remediation plan to open the **Plan details and status** card on the **General tab**. The latest execution status of the remediation plan is displayed.



NOTE

A status of **N/A** indicates that the remediation plan has not yet been executed from within Red Hat Lightspeed.

4. **Optional:** Go to the **Execution History** tab to view status or progress of the execution on each system included in the plan, with links to detailed execution activity logs.

Next steps

To monitor the status of a playbook in the Satellite web UI, see Executing a Remote Job in the Red Hat Satellite *Managing Hosts* guide.

5.8. ADDITIONAL RESOURCES

- Creating a remediation plan in Lightspeed

CHAPTER 6. EXECUTE REMEDIATION PLANS

After you create a remediation plan, you can execute the playbook on remote systems from the Red Hat Lightspeed application. You can also download and run the playbook by using your organization's Ansible Automation Platform (AAP) workflow.

6.1. REMEDIATION PLAN EXECUTION READINESS

When you create a remediation plan, you must ensure that the plan can successfully execute on your systems. Before a plan can execute successfully, it must meet the criteria for execution readiness in the Hybrid Cloud Console.

As a general rule, you can run smaller-scale remediation plans directly within Red Hat Lightspeed. Use Red Hat Lightspeed for rapid, targeted remediations.

A larger-scale remediation plan contains large numbers of systems or includes multiple complex remediation operations, or both. If a larger-scale plan cannot execute directly on Red Hat Lightspeed, you can download it to run with Ansible Automation Platform (AAP). AAP allows you to execute at scale with greater functionality designed specifically for enterprise-grade execution, such as advanced orchestration capabilities: scheduling, RBAC, and auditing features.

You can also download a larger-scale remediation plan and use a Red Hat Satellite integration to execute it.

Additional resources

- Downloading remediation plans
- Executing remediation plans
- Setting up Red Hat Insights for Red Hat Ansible Automation Platform Remediations
- Creating an Insights remediation plan for hosts

6.1.1. View execution readiness criteria

Execution readiness criteria are conditions that your remediation plan must meet before the plan is ready to execute. Before you execute a remediation plan, ensure that your plan meets the criteria.

A remediation plan must meet the following criteria before it can execute:

- You have the required user permissions to run the remediation plan on your systems.
- The remote host configuration manager is enabled.
- The rhc client is active on all systems.
- All systems are connected and visible in Red Hat Lightspeed inventory.

Remediation plans must fall within *execution limits* before you can execute them remotely by using Red Hat Lightspeed. An execution limit is the threshold that determines whether a remediation plan can successfully execute on your systems.

If a remediation plan falls outside execution limits, you cannot execute the plan within Red Hat Lightspeed. However, you can download the plan and run it manually with Ansible Automation Platform or with a Red Hat Satellite integration.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console.
- You have the required user permissions to run remediation plans on your system.

Procedure

1. Navigate to Automation Toolkit > Remediation Plans in the Red Hat Hybrid Cloud Console.
2. Select the remediation plan that you want to view from the list. The details page for the plan displays.
   The **General** tab shows details about your remediation plan, such as creation date and number of systems in the plan. The **Execution Readiness Summary** lists the readiness criteria for your plan, even if the plan does not meet the criteria.

   If your remediation plan meets the criteria, the summary shows checkmarks next to each of the criteria, and the status message shows **Ready**. The **Execute** button is available.

   If your plan does not meet the readiness criteria, the **Execute** button is disabled and the status message **Not ready (# errors)** displays. You can modify the plan to run with Red Hat Lightspeed, or download it and execute it with Ansible Automation Platform or a Satellite integration.

Additional resources

- Integrate with Red Hat Lightspeed (formerly Insights)
- Configure automation controller
- Monitoring hosts by using Red Hat Insights

6.1.2. About action points

Red Hat Lightspeed uses action points to calculate the relative complexity and performance impact of a remediation plan. Complex issues require a higher number of action points, and simpler issues require a lower number. Red Hat Lightspeed assigns a specific point value to each type of issue and calculates the plan's total size against the platform's performance capabilities.

Red Hat Lightspeed supports execution reliability for plans up to 100 systems and 1,000 action points. If the total number of systems and action points for your plan falls within these limits, you can run the plan directly on Red Hat Lightspeed.

Plans that exceed these limits cannot be executed directly in Red Hat Lightspeed. The **Execute** button is disabled, and the Execution readiness summary indicates how many systems and action points to remove. Alternatively, you can download the plan and run it with Ansible Automation Platform (AAP) or a Satellite integration.

Additional resources

- Creating an Insights remediation plan for hosts
- Executing remediation plans
- Setting up Red Hat Insights for Red Hat Ansible Automation Platform Remediations

6.2. OPTIMIZE THE NUMBER OF PLANNED REMEDIATION LIMITS

The **Execution readiness summary** shows the number of systems and action points in your remediation plan. If the plan exceeds 100 systems and 1,000 action points, the summary indicates how many to remove before the plan can run in Red Hat Lightspeed.

If your plan is slightly over the execution limits, you can remove individual systems or issues from the plan to make reductions in your planned remediation. Removing a single advisor issue frees up 20 points, whereas you would need to remove 10 Patch issues to achieve the same reduction.

Use these guidelines to view the number of action points required to run your remediation plan:

- Advisor issues: 20 points for each issue (High complexity)
- Vulnerability issues: 20 points for each issue (High complexity)
- Compliance issues: 5 points for each issue (Medium complexity)
- Patch issues: 2 points for each issue (Low complexity)

For example, a plan containing 50 Patch issues would only total 100 points (50 x 2), which falls well within execution limits. However, a plan with 50 advisor issues would total 1,000 points (50 x 20), reaching the maximum limit immediately. The plan with 50 advisor issues would still run on Red Hat Lightspeed, but only if the number of systems is fewer than 100.

If your plan falls slightly over the limit of 1,000 action points, you can remove individual issues from the plan to make reductions in your planned remediation. Removing a single advisor issue frees up 20 points; you would need to remove 10 patch issues to achieve the same reduction.

If your plan significantly exceeds the limits, you can download it and execute it with Ansible Automation Platform instead of removing systems or issues.

Additional resources

- Modifying the actions in a remediation plan
- Removing a remedial action from a remediation plan
- Removing a system from a remediation plan

6.3. EXECUTE REMEDIATION PLANS FROM THE RED HAT LIGHTSPEED UI

You can execute the playbooks generated by your remediation plans from the Red Hat Lightspeed UI on the Red Hat Hybrid Cloud Console, if you have the required permissions and pass the readiness check.

Execution readiness check

When you open a remediation plan in the Red Hat Lightspeed UI, a readiness check is automatically activated. The readiness check ensures that you have completed the prerequisites necessary to execute the playbook on the systems that are included in the plan.

The results of the readiness check display in the **Execution readiness** section of the remediation plan details, along with some hints to help you understand which prerequisites were not met.

If the readiness check fails, the **Execute** button is disabled. This does not necessarily mean that there is an issue with your plan. It could mean that there is an issue connecting to the remote host or that you do not have the required permissions to connect and trigger the execution readiness checks.

Required permissions for remediation plan execution

You need either the **Remediations administrator** role or the **RHEL administrator** predefined User Access role to execute a remediation plan on remote hosts from Red Hat Lightspeed.
The predefined roles are managed and granted by your **Organization Administrator** in the Identity and Access Management (IAM) settings on the Red Hat Hybrid Cloud Console.



IMPORTANT

- The **Remediations user** and **RHEL operator** predefined roles can create and edit remediation plans but cannot execute them on remote systems. The **Remediations administrator** or **RHEL administrator** role grants full remediations access and system connection discovery.
- Without the required permissions, the connection status for your system displays as **Unknown**, even if you can connect to that system.
- The **Remediations administrator** or **RHEL administrator** predefined User Access roles are not default roles. An Organization Administrator will need to create the group, assign one of these roles, and then add the specified user to the group. For more information about User Access permissions, see Managing group access with roles and members.

Prerequisites

- You are logged in to the **Red Hat Hybrid Cloud Console**.
- Your user account is a member of a User Access group with one of the following predefined roles:
  - **Remediations administrator**
  - **RHEL administrator**
- You have completed the steps in Enabling host communication with Red Hat Lightspeed
- The option **Allow permitted Red Hat Lightspeed users to execute remediation playbooks on rhc-connected systems** is enabled on the Remote Host Configuration Manager page in the Red Hat Lightspeed UI.
- You have the **Organization Administrator** or **Remediations administrator** role.
  IMPORTANT:
  - The **Remediations user** role cannot execute remediation plans on remote systems. The **Remediations administrator** role grants full remediations access and system connection discovery.
  - Without the required permissions, the connection status for your system displays as **Unknown**, even if the system is otherwise accessible.

    - The **Remediations administrator** role is not a default role. You must create the group and add yourself. For more information about User Access permissions, see Managing group access with roles and members.

- You have successfully completed the execution readiness check in the remediation plan in the Red Hat Lightspeed UI. You can find the readiness results in the **Execution readiness** section of the remediation plan details.

Procedure

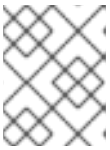
1. In the Red Hat Hybrid Cloud Console, navigate to Automation Toolkit > Remediation Plans.
2. Scroll through the list and find a remediation plan.
3. Click the name to open the **Remediation plan** details view.
4. Click **Execute**.



IMPORTANT

If the **Execute** button is disabled, this means that the execution readiness check failed because one of the requirements was not met. To help you troubleshoot and complete the execution readiness check successfully, see Execution readiness check.

5. When prompted, click **Execute playbook on systems** The playbook runs on the systems included in the remediation plan.



NOTE

A remediation plan with a large number of actions to execute on many systems might take a while to complete.

Next steps

- To monitor the progress, go to the **Execution History** tab for the plan you executed. The **Execution History** tab displays the status, history and links to the logs of a plan execution for each included system.
- After the remediation plan executes successfully, open the recommendation or issue it addressed. Verify that the remediated systems no longer appear in the list.

Additional resources

- Registering RHEL systems and configuring client tools with Red Hat Lightspeed
- User Access configuration guide for role-based access control (RBAC)

6.4. EXECUTE REMEDIATIONS FROM THE SATELLITE UI

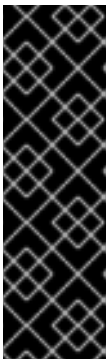
You can also remediate using the Satellite UI.

Prerequisites

- You are a **Cloud Administrator**.
- Your user account is a member of a group that has access to one of the following predefined User Access roles:
  - **Remediations administrator**
  - **RHEL administrator**
- You have completed Host registration by using the insights-client.

Procedure

1. To remediate a recommendation on RHEL systems managed by Red Hat Satellite, see Remediating issues based on Red Hat Lightspeed recommendations in the Satellite *Managing hosts* documentation.



IMPORTANT

When you introduce a new host into your Satellite inventory, by means of provisioning or registration, two automatic background tasks initiate. It takes 24 hours for these tasks to complete. This is a typical time frame for the automatic synchronization.

If you identify security issues or another scenario that warrants not waiting 24 hours for the automatic sync, you can manually synchronize by clicking the sync button in the UI. This manual sync completes in a few minutes.

2. To view the procedures for enabling automatic and manual synchronization, see the Configuring Synchronization of Red Hat Lightspeed Recommendations for Hosts in the Satellite documentation.

CHAPTER 7. MANAGE REMEDIATIONS IN RED HAT LIGHTSPEED--ASSOCIATED PLAYBOOKS

You can create, configure, and manage Red Hat Lightspeed remediation plans and their associated playbooks.

7.1. INSTALL THE INSIGHTS-CLIENT ON SATELLITE SERVER CONTENT HOSTS

The insights-client comes preinstalled on most versions of Red Hat Enterprise Linux. If the client is not present, you can install the insights-client on each system.

Prerequisites

- Register your hosts to Satellite
  - If you already have Red Hat Enterprise Linux hosts, you can use the Global Registration Template to register them to Satellite. For more information, see Registering Hosts in the Satellite documentation.

Procedure

1. Install the insights-client:

```
# yum install insights-client
```

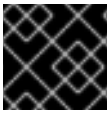
2. Register the host to Red Hat Lightspeed:

```
# insights-client --register
```

3. Repeat these steps on each host.
4. Alternatively, you can use the **RedHatInsights.insights-client`** Ansible role to install the insights-client and register the hosts. For more information, see Monitoring hosts by using Red Hat Lightspeed in the Red Hat Satellite *Managing Hosts* guide.

7.2. CONFIGURE CLOUD CONNECTOR AFTER UPGRADING SATELLITE SERVER 6.10 TO 6.11

After you upgrade your Satellite Server, configure Cloud Connector to complete the upgrade process.



IMPORTANT

The following information applies to upgrades from Satellite Server version 6.10 to 6.11.

For more information, see Upgrading and Updating Red Hat Satellite .

Procedure

- To configure Cloud Connector after upgrading the Satellite Server, click **Configure Cloud Connector** from **Configure → RH Cloud - Inventory Upload**to enable it on the new version of

Satellite Server. Simultaneously, you must remove the previous source from the cloud manually on the Red Hat Hybrid Cloud Console after upgrading your Satellite Server.

Verification

- After you configure the Cloud Connector, it removes the receptor components and installs the RHC components. At the same time, the Cloud Connector announces all the organizations in the Satellite to the source and is ready to receive the connections.

7.3. DISABLE DIRECT REMEDIATIONS ON A SATELLITE SERVER CONTENT HOST

You can disable direct remediations to ensure the playbook run does not get invoked from the cloud on a single host.

The **enable_cloud_remediations** setting in Satellite controls whether your hosts can be directly remediated through the Red Hat Lightspeed console. When the value of **enable_cloud_remediations** is set to **False**, direct remediation from Red Hat Lightspeed is disabled for that specific host. The parameter is not set for your systems, which means that by default, direct remediation is enabled with Cloud Connector.



NOTE

All hosts in the organization inherit the value that is set for **enable_cloud_remediations**.

For information about how to enable automatic and manual synchronization, see the Configuring Synchronization of Red Hat Lightspeed Recommendations for Hosts in the Satellite documentation.

When Satellite receives the request to run the remediation plan's playbook from Cloud Connector, the request includes a list of hosts on which to execute the playbook.

Procedure

1. Go to **Hosts menu → All Hosts** in the Satellite web UI.
2. Locate the host, and click **Edit → Parameters**.
3. Set the **enable_cloud_remediations** parameter to **False**.

7.4. DISABLE DIRECT REMEDIATION ON A SATELLITE SERVER CONTENT HOST GROUP

You can disable remediations for the entire organization or host group by changing the **Global Parameters** in the Red Hat Satellite UI.

Prerequisites

- You have an **Organization Administrator** role.

Procedure

1. Navigate to the Satellite Dashboard.

2. Click **Configure → Global Parameters → Create Parameter**.
3. In the **Name** field, enter **enable_cloud_remediations.**
4. In the **Value** field, enter **false**.
5. Click **Submit**.

Verification

- Check to see whether your new parameter is listed in the **Global Parameters** table. For more information about configuring global parameters, see the Parameters section of the Red Hat Satellite documentation.